

A Novel Method and Testbed for Sensor Management and Patient Diagnosis *

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Abstract

Low-cost sensors and wireless systems can now create a constantly vigilant and pervasive monitoring capability at home, at work, and in conventional point-of-care environments. While progress in this area is underway in sensor technology, mobile computing platforms, and data transport, barriers to large scale application remain ahead, particularly in the area of patient disease diagnosis, which generally requires a diverse set of sensors and instruments that are applied at proper times in response to patient state/behavior. As these sensors may be numerous, and may not be worn comfortably and practicably at all times, a solution is required for the systematic selection of sensors at the point of use. We describe the Incremental Diagnosis Method (IDM) system, an embedded decision support system based on Bayesian statistics and decision analysis theory developed to select or deselect available sensors so that the diagnostic certainty of patient condition best improved while the set of sensors used on the patient body is minimized. IDM has been evaluated in a testbed, the Medical Embedded Device for Individualized Care (MEDIC) system, based on standard, ubiquitous wireless platforms. MEDIC supports local sensing and signal processing, autonomous decision support, and remote reconfiguration and control of wearable components. A detailed evaluation of IDM operation and performance for patient gait analysis is also given in this paper. Finally, we also discuss the many new opportunities provided by IDM and the related future research introduced by this capability.

1. Introduction

The rapid advances in patient support and wellness healthcare programs have created requirements for new data sources provided by biomedical sensor monitoring of patients in their home- and work-life. This also introduces a new paradigm for monitoring at the point-of-care requiring the patient information be readily available to any physician at location engaged in the patient's treatment. Finally, this also emphasizes the need for enabling involvement by patients in their own monitoring. Promising results of investigations in a range of medical disciplines combined with progress in wireless consumer electronic platforms indicate the potential for a transformation in patient care based on the augmentation of standard clinical methods with new biomedical, wearable sensor data sources.

The current proliferation of broadband wireless services, along with more powerful and convenient handheld devices, will enable the introduction of real-time monitoring and guidance for a wide array of patients. Low-cost sensors and wireless systems can now create a constantly vigilant and pervasive monitoring capability at home, at work, and in conventional point-of-care environments (e.g., primary care physician offices, outpatient clinics, rehabilitation centers). Indeed, a large research community and a nascent industry is beginning to connect medical care with technology developers, vendors of wireless and sensing hardware systems, network service providers, and enterprise data management communities. Wearable devices focusing on personal health, rehabilitation, and early disease detection are now being prototyped. While progress in this area is underway in sensor technology, mobile computing platforms, and data transport, barriers to large scale application remain ahead. A lengthy research program combining wireless biomedical monitoring with the discipline of Medical

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Informatics, has shown that the primary objective of patient disease diagnosis generally requires a diverse set of sensors and instruments that are applied at proper times corresponding to environment, patient behavior, and inferred patient condition. Since these sensors may be numerous, and may not be worn comfortably and practicably at all times, a solution is required for the systematic selection of sensors at the point of use.

We have developed the Incremental Diagnosis Method (IDM) [79], an embedded decision support system, which provides user guidance for the selection of sensors based on direct computation of both the resource cost and the contribution of each sensor to the certainty of diagnosis. Based on Bayesian statistics and decision analysis theory, IDM provides a framework to discern when to use available sensors to improve diagnostic certainty at various resolution levels of patient condition. The IDM algorithm views the set of sensors and instruments in a hierarchical manner, and provides a method for determining from this organization which sensors must be (de)activated to best answer specific questions. Starting with a minimum set of low-cost, full-time use (i.e., constantly activated and vigilant) wearable sensors, IDM gradually adjusts to higher-cost, part-time use (e.g., periodically worn and activated) sensors at the appropriate times. In each case, fusing of patient data is performed so that a particular medical condition can be inferred with an acceptable level of certainty. The use of the IDM applies to wearable sensors and standalone instruments as well as other clinical data sources. This paper describes an autonomous IDM solution being adopted for gait analysis in rehabilitative medicine, as well as the broad range of research opportunities created by this general method and the shared data resources it provides.

While IDM system design is developed for application to many wireless and mobile computing technology choices, we implemented IDM in a testbed based on standard, ubiquitous wireless platforms. This paper also describes the testbed that implements IDM and can be applied to a variety of biomedical scenarios. This testbed, Medical Embedded Device for Individualized Care (MEDIC) [78], a new hardware and software architecture based on standard commercial systems that support local sensing and signal processing, autonomous decision support, and remote reconfiguration and control of the system. Combining IDM as a client application on a MEDIC testbed, we show how a wireless medical system can monitor patient health, while at the same time, provide real-time and continuous disease diagnostics assistance. We would like to emphasize that the testbed presented here, along with IDM systems are open source shared contributions, available to the scientific community for further investigation and development, thus encouraging community collaborations to advance both technology and applications.

2 Background

The classical “hospital centered healthcare” model calls for a wearable system that fills in the “gaps” in health monitoring between clinic visits, and relies on medical personnel at the centralized location for analysis of the data. This requirement results in a wearable system that simply forwards raw physiologic data to a centralized location for offline processing [17]. As the focus of health monitoring shifts from the “hospital-centered” to “patient-centered”, the wearable systems evolve from the simple data acquisition devices to intelligent systems capable of making local medical diagnosis leading to the timely and most appropriate medical intervention [54]. The intelligent wearable systems may decide upon the most appropriate treatment alternative for a specific patient, from choosing among different sensors for further diagnosis to prescribing therapeutic regimes. Thus, the information processing in the intelligent wearable systems capable of making local treatment decisions involves two steps: 1) determination of patient’s medical condition, and 2) reasoning about the effects to be expected from different treatment alternatives [50].

In determining the patient’s medical condition, some wearable systems focus on monitoring a single dominant physiological feature as a symptom of a medical condition by performing a simple rule-based classification on individual sensor data to generate alerts [55, 22]. Modeling a medical problem with a single physiologic sensor, however, only gives a first-approximation of the medical condition and may not lead to a powerful medical diagnosis as when combined with other physiologic sensors and the patient’s context. The contextual information describing the patient’s activity, environment, and social interaction can enhance the diagnostic power of a wearable sensor system [19, 73, 75] because of the correlations between the physiological and contextual data [16]. For example, additional knowledge from activity sensors (e.g., accelerometers) showing that the patient is rigorously exercising (e.g., in an aerobic state) can discount the fact that the patient may be suffering from a heart problem when increased heart rate is observed on the ECG sensor. Here, the user context comprises the user’s static posture, dynamic activity, orientation, gestures, biometric, and emotional states. The location, time, humidity, barometric pressure, illumination, and noise characterize the environmental context [33]. The social interaction includes contact and communication with people nearby [53, 72]. In addition, combining different sensing modalities conveys more information about the patient’s state by leveraging the inherent strengths of different sensor types [24].

The subsequent sections describe the statistical learning techniques that are available for determination of patient’s medical condition, and the influence diagram formalism to

select an optimal treatment based on the predictions.

2.1 Statistical Learning

Patterns in the physiologic and motion signals can be processed with pattern recognition methods used widely in statistical learning [34, 25, 43], which can be divided into feature extraction and classification. In feature extraction step, features that are indicative of different patient states (patterns) are chosen from the sensor measurement [26, 33]. The features can be temporal: dynamic range ($max-min$), mean, standard deviation, period, rate variation, correlation between different channels, entropy, and signal morphology [64], and/or spectral: energy, power spectral density, moments, entropy, wavelet, and eigen components [61]. The temporal features achieves high classification accuracy at low sampling rates while spectral features work better at higher sampling rates [33, 44]. Only features that are relevant to the classification problem should be selected [14, 81].

Neural networks [70], decision trees [69], k-nearest neighbors [18], and Bayesian networks [67] are well known classification techniques that can be used to recognize the patterns in medical diagnosis through the use of knowledge bases inductively learned from previous cases [40]. Adaptive and learning properties of neural networks are used to recognize repetitive patterns of the patient states [59]. Neural networks, however, are typically used as black-box classifiers lacking transparency in knowledge representation and in general cannot easily explain their decisions [49, 41]. Decision tree methods can classify classes and subclasses of patient's state at different hierarchical levels [57, 66, 38]. The decision trees, however, have properties that may not be suitable for use in medical diagnosis [40]. The trees are generally pruned to produce general rules, usually resulting in an uninformative, short tree with a poor description of the diagnosis. When classifying with a decision tree, only a few attributes along the path from the root to the leaf is examined, even if the values of other attributes are available. The decision trees are also sensitive to missing data; the classification becomes unreliable if the value for some attribute especially near the root of the tree is missing. The k-nearest neighbors (kNN) approach associates observations with the closest patient context [46] with an assumption that observations which are close together – for some appropriate metric – will have the same classification. The kNN method, however, cannot explain the decision of the algorithm [41].

In contrast, the Bayes theory offers a fundamental approach to the pattern classification tasks in medical diagnosis [40, 49]. Bayesian theory is based on the conditional probability that can be used to model the causal relationships between the human context and sensors features, ob-

jectively inferring the hidden contextual information from the observed sensor data [67]. Bayesian theory also allows the description of the uncertainties present in the sensors by means of probability distributions [56]. In addition, this approach can fuse sensor data by representing the various sensor data via a joint distribution. Bayes theory is based on coherent statistical methods that are fundamentally compatible with scientific approach. Bayesian methods make it possible to incorporate a scientific hypothesis in the analysis (by means of prior distributions), and provide the basis to update the original set of beliefs in a hypothesis in light of new information from the observed data, producing a more refined set of beliefs [13]. This transparency and explanation ability of the Bayesian theory has been argued to be similar to the physicians' way of making decisions [40]. Thus, Bayesian theory is especially suited for reasoning with the inherent dependencies and uncertainties in medical models that are involved in establishing diagnoses, selecting treatment alternatives, and predicting treatment outcomes [50].

Bayesian methods are implemented using probabilistic graphical models known as Bayesian belief networks (BBN). BBNs are effective in medical diagnoses [50], but can involve thousands of variables and require computing power that is still out of the reach of wearable systems. A special case of BBNs, known as naïve Bayes, are more suitable and scale well for wearable sensor systems that have limited computing capability as naïve Bayesian networks can be implemented with efficient algorithms for induction while maintaining high robustness to irrelevant attributes [47, 40]. Like the general BBNs, naïve Bayes networks can be constructed from probability methods, however, with an assumption that attributes are conditionally independent if the class is known, a premise typically true in medical datasets [41]. An advantage of the naïve Bayes networks is that the parameters, which correspond to the initial set of beliefs about the relative plausibility of various hypotheses, can be estimated experimentally and efficiently from the observed data by the maximum likelihood method [40]. Bayes classifiers are effective in motion recognition applications [39, 37, 43], as well as in many other applications such as text classification, medical diagnosis, and system performance management [20, 74, 27], often outperforming more sophisticated techniques [21, 20, 23, 29, 47].

2.2 Decision Making

The patient's condition classified in a medical diagnosis needs to be transformed into actions in choosing various treatment alternatives [71]. The formalism of the classification techniques in statistical learning do not provide reasoning of the effects to be expected from different treatment

alternatives. The constructs from the decision theory can extend the classifiers such as Bayesian networks to include knowledge about decisions and preferences [50]. This extension permits an objective classification of patient's condition while subjective preferences are incorporated in deciding various treatment alternatives.

The treatment alternatives can be decided locally by using utility functions, which define the preference relationship between different treatments given the predictions of a given set of medical conditions, and give the users (the medical professional and the patient) the opportunity to rationalize their choices. Because the medical conditions are determined probabilistically using statistical learning techniques, the choice of treatments corresponds to decision making under uncertainty [76, 80, 15]. This user-defined utility function can be used as the framework to rationalize the trade-off between higher resolution levels, detection certainty, and user cost and convenience in intelligently selecting which sensors to be included in performing the diagnosis.

2.3 Wearable Sensors and Medical Devices

Recent research has focused on mobile health monitoring based on new miniaturized sensors (e.g., micro-electro-mechanical systems (MEMS)) and widespread adoption of wireless networks [52]. A plethora of sensors are available to measure motion (e.g., accelerometers), physiology, and other environmental variables that reflect an individual's state, enabling data capture and access in point-of-care environments. A multitude of different devices have been developed allowing a patient to wear a set of sensors. Common designs include: wristbands for measuring pulse, body temperature, galvanic skin reactions, and electromyography (EMG) data [63, 68, 17, 9]; chest and arm belts for physiologic monitoring [51, 35, 45, 31, 60, 22]; shoes for gait analysis [58, 65]; and photoplethysmographic ring sensors [11]. It is important to note that many of these systems primarily focus on physiological data acquisition, and do not provide methods that assist healthcare providers (or patients) with data interpretation or diagnosis.

At the same time, advancements in wireless technology enable a 3-tier network architecture, which replaces some of the cables between the personal server and individual sensors with a wireless body area network (WBAN) [32, 36]. Due to the close proximity of the wearable sensors in the WBAN, low-power short-range radio standards such as Bluetooth and Zigbee can be used to link the wearable sensors and the processing unit [63, 36]. Recognizing that it is not always plausible for a sensor to be worn, the same wireless technology can be used to connect home instruments with the wearable systems [30, 42, 77, 28]. Non-portable

sensing devices (e.g., weight scales, cameras) are equipped with additional wireless network capabilities to relay captured data. The WBAN offers improved system wearability at higher system power consumption [10]. An extension of WBAN in the third tier is the use of wearable sensors in an ad-hoc network that can be deployed at a mass-casualty site in an emergency situation [48].

Local signal processing and decision making for sensor selection are currently not present on these systems, and several challenges still exist in understanding the requirements for WBANs in healthcare. Specifically, we note that mote technology (e.g., as used in CodeBlue and related projects [55, 62]) has provided important lessons related to the development of biomedical monitoring capabilities. However, mote platforms lack the memory, storage, and processor capabilities for supporting multiple applications with local signal processing, sensor data storage, data processing, and finally user interaction. Fortunately, the progress in microelectronics that once enabled the first motes has now continued to create ubiquitous and capable cellular phone embedded computing platforms operating with standard wireless interfaces. And now, compact, low-power and long-lifetime sensors may accompany these devices as body worn systems.

3 Incremental Diagnosis Method

The Incremental Diagnosis Method (IDM) is an iterative process with an ability to adjust sensors to achieve the diagnostic resolution level and certainty requirements. The proposed method minimizes the set of activated sensors by choosing only appropriate sensors as needed.

3.1 Diagnostic Resolution Levels

The level of resolution is related to appropriate configuration of different types of sensors that are required to identify a medical condition with high certainty. When diagnosing a medical condition, successively more refined levels of information are typically acquired. The concept of diagnostic resolution levels can be envisioned as a tree structure, in which the root corresponds to the least informative state and the leaves describe the patient state in detail. As such, the root state may only require a low-resolution sensor for detection, while the leaf states typically require high-resolution sensors. These data sources that can better describe patient states may be considered costly (e.g., expensive, inconvenient, high power usage, etc.), and hence, may not be desired in the wearable system when not needed.

As an example, diagnostic resolution levels for patient ambulation leading to the detail limp detection are shown in Fig. 1. The top most level corresponds to the basic ambulation states (walking, limp, limping, etc.) and requires

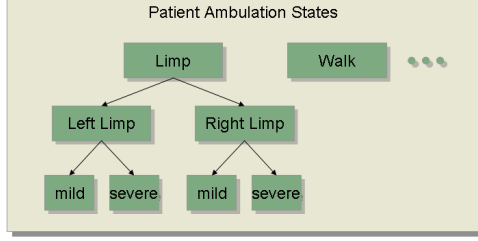


Figure 1. A three-level resolution diagram of a patient ambulation describing limp as an example. The resolution levels correspond to varying degrees of information about the limp.

a minimum number of sensors to detect; a single triaxial accelerometer has been shown to be sufficient for this query [43, 37]. Information regarding a more specific limp condition (e.g., *Is the limp on the right leg?*) can be important for medical diagnosis (e.g., a patient with edema on the left side of the brain). Two triaxial accelerometers attached to each side of the hip may be required for this query. Yet a more refined state of the patient limp (e.g., *How severe is the limp?*) may also be of interest. Such a detailed query requires fusing more information from a knee angulation sensor that may be inconvenient for the patient to wear.

3.2 Algorithm Overview

The IDM algorithm (Fig. 2) has three inputs: (1) a *diagnostic resolution level* query, (2) a minimum acceptable *certainty threshold* for detection, and (3) a *user action* that results in acquisition of new sensor inputs. Available sensors can include a set of sensors ranging from full-time (e.g., an accelerometer) and part-time (e.g., a knee sensor) wearable sensors to standalone instruments such as a weight scale. The wearable system continuously acquires data from available sensors (i.e., all full-time and some part-time wearable sensors that are present) and infers the user state (e.g., ambulation) using the real-time inference engine (Section 3.3). The system outputs are a report of the patient state, and sensor requests.

There are two feedback loops in the system. The first one occurs when the system has control of the part-time sensors, but does not always include these sensors in the analysis to conserve system energy. If needed, the IDM could activate these sensors automatically to acquire data for the next diagnostic iteration. The second loop involves a user activating the part-time sensors or data acquisition from standalone instruments. This loop allows the utilization of sensors and instruments (e.g., blood-pressure monitor, weight scale, knee sensors, etc.) that are inconvenient or impossible to wear

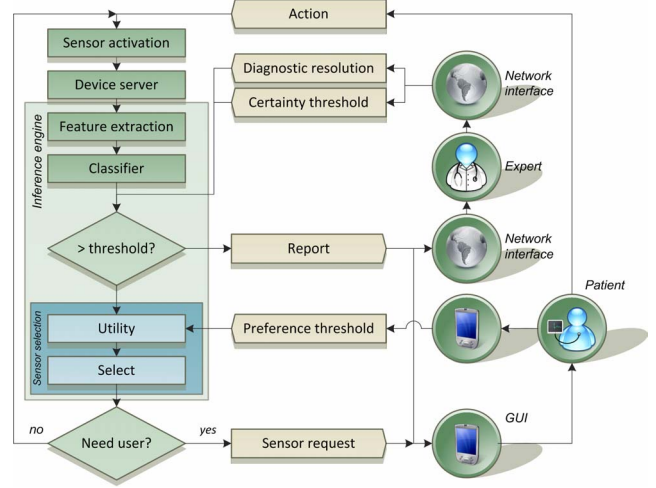


Figure 2. Architecture overview of the Incremental Diagnosis Method.

full-time, though occasional use can greatly enhance detection of a particular user state. The feedback loops terminate when the desired resolution level and certainty threshold are achieved, or the available sensor set is exhausted but the inference certainty is still below the desired threshold. In the latter case, training of a new model, possibly with additional sensors, should be revisited. The modular architecture permits new sensors and external models to be seamlessly incorporated into the algorithm.

3.3 Inference Engine

The inference engine, which is hosted on the wearable system, consists of three components: (1) *Feature Extraction*, (2) *Naïve Bayes Classifier*, and (3) *Sensor Selection*. Each component is described next.

3.3.1 Feature Extraction

The feature extraction step summarizes the time domain data from each sensor into a vector of sensor feature variables $\mathbf{F} = \{F_1, \dots, F_n\}$ for the Bayes classifier (see below). For patient ambulation in steady states, sensor readings in the time domain can be processed to extract frequency spectral components by Fourier transform. The fundamental (single dominant) frequency component and spectrum energy have been shown to be important features for physical activity recognition [12]. In addition, the efficient computation of Fast Fourier Transform (FFT) is attractive for use in resource-constrained wearable systems. Note that in the general case, features in the time domain can also be vital in characterizing other patient states [33]. Due to

the modular architecture, additional feature extraction procedures can be incorporated in the inference engine without changes to the other parts of the system.

3.3.2 Naïve Bayes Classifier

In this step, the inference engine uses a naïve Bayes classifier model [67] to infer patient state C probabilities given the feature vector $\mathbf{F}$ from the previous step. Due to the “naïve” assumption – if the patient state C is known, each feature F_i is conditionally independent of every other feature F_j for $j \neq i$ – the conditional probability of the feature vector given the patient state can be expressed as a simple product of the conditional probabilities of each feature variable given the patient state, i.e., $p(\mathbf{F}|C) = \prod_{i=1}^n p(F_i|C)$. Likewise, the marginal probability of the feature vector can be expressed a simple product of the prior marginal probabilities of each feature, i.e., $p(\mathbf{F}) = \prod_{i=1}^n p(F_i)$. As such, the conditional distribution of the patient states can be expressed as shown below.

$$p(C|\mathbf{F}) = \frac{p(C) \prod_{i=1}^n p(F_i|C)}{\prod_{i=1}^n p(F_i)} \quad (1)$$

As a result of this step, one can infer the probabilities for the patient being in one of the states C given the sensor feature vector extracted in the previous step (Section 3.3.1). Note that such a simple expression (Eq. 1) makes the naïve Bayes classifiers very attractive in resource constrained applications, as the parameters in the equation can be calculated efficiently using supervised training [79].

3.3.3 Sensor Selection

In the final step, the inference engine determines if the diagnosis meets the required level of accuracy, as required at the outset by the user. If not, a decision is made as to which of the higher-cost sensors to include in the next processing iteration. A utility function guides the suggestion of actions that are recommended to the user, such as inserting/removing part-time wearable sensors, or the use of a standalone instrument. The utility function consists of three parts: (1) the formulation of a decision matrix D that denotes a simple form of an expert opinion and user preferences in sensor selection, (2) the calculation of the utility of each action, and (3) the selection of the action that yields the most utility. In the $[m \times n]$ decision matrix D , shown below, the columns correspond to the patient states, z_j , and the rows correspond to a decision in selecting alternative sensor configuration, a_i . The parameter d_{ij} defines a relative influence of z_j in choosing a_i .

$$D = \begin{matrix} & z_1 & \dots & z_j & \dots & z_m \\ \begin{matrix} a_1 \\ \dots \\ a_i \\ \dots \\ a_n \end{matrix} & \begin{pmatrix} d_{11} & \dots & d_{1j} & \dots & d_{1m} \\ \dots & \dots & \dots & \dots & \dots \\ d_{i1} & \dots & d_{ij} & \dots & d_{im} \\ \dots & \dots & \dots & \dots & \dots \\ d_{n1} & \dots & d_{nj} & \dots & d_{nm} \end{pmatrix} \end{matrix}$$

In the formulation of this decision matrix, an expert opinion can narrow down the problem of estimating d_{ij} . For example, an expert may determine the use of some sensors to be independent of some patient states by assigning zero (0) to d_{ij} , to indicating that z_j has no influence over a_i . The relative values of the non-zero parameter d_{ij} defines the user’s preferences in choosing one out of a set of n alternatives in the presence of m states. The values are chosen so that the ratio (Eq. 2) between the expected values of the relevant (non-zero) rows correspond to the user preference threshold $\theta_{i,k}$ in selecting different alternatives a_i and a_k .

$$\theta_{i,k} = \frac{EU(a_k)}{EU(a_i)} \quad (2)$$

The *expected utility* $EU(a_i)$ of each alternative configuration is calculated by incorporating the patient state:

$$EU(a_i) = \sum_{j=1}^m p_j d_{ij} \quad (3)$$

where $p_j = p(C = c_j|\mathbf{F}) \geq 0$ and $\sum_{j=1}^m p_j = 1$, a probability distribution obtained from the naïve Bayes classifier in Eq. 1. Then, the alternative with the largest expected utility is selected as the optimal sensor configuration:

$$i^* = \operatorname{argmax}_{i=1,\dots,n} EU(a_i) \quad (4)$$

where the function $\operatorname{argmax}_x[f(x)]$ returns an argument x such that the value of the function $f(x)$ is maximized. It is important to note that each alternative must be optimal in at least one of the patient states. Otherwise, the alternative is said to be dominated and can be removed from the decision matrix [15].

As an example, consider a case involving six states and four decisions – choosing either of the two optional sensors (A or B), their combination (A and B), or no sensors. Thus, the alternatives are $\{a_1=A, a_2=B, a_3=AB, a_4=\text{none}\}$. Assume also that an expert states that only z_1 and z_2 have influence on choosing a_1 ; only z_3 and z_4 have influence on choosing a_2 ; only z_5 and z_6 have influence on choosing a_4 ; and that all of the states have influence on choosing a_3 . The decision matrix for this example, D_{ex} , can be formulated as shown below:

$$D_{ex} = \begin{matrix} & z_1 & z_2 & z_3 & z_4 & z_5 & z_6 \\ \begin{matrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{matrix} & \begin{pmatrix} d_{11} & d_{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & d_{23} & d_{24} & 0 & 0 \\ d_{31} & d_{32} & d_{33} & d_{34} & d_{35} & d_{36} \\ 0 & 0 & 0 & 0 & d_{45} & d_{46} \end{pmatrix} \end{matrix}$$

The user preference threshold between selecting a_1 and a_3 is the ratio:

$$\theta_{1,3} = \frac{E(d_{3j})}{E(d_{1j})} \quad (5)$$

where $d_{ij} \neq 0$. In this case, the utility function will select a_1 when $p_1 + p_2 > \theta_{1,3}$; p_1 and p_2 are probability of states z_1 and z_2 , respectively. Otherwise, a_3 will be selected as the decision. The parameters for the decision matrix can be selected by leveraging domain knowledge and patient preferences in sensor usage [79].

4 MEDIC

A proof-of-concept, end-to-end prototype of a system, MEDIC (Medical Embedded Devices for Individualized Care), was developed with a suite of Bluetooth-enabled wearable sensors (blood oximeter, accelerometers), personal instruments (blood pressure cuff, weight scale), and network interfaces [78]. In addition to Bluetooth for wireless body area networking (WBAN) of biometric sensors and instruments, MEDIC handheld supports WiFi for connection to backend servers at hospitals and clinics. This initial implementation supports sampling and event detection, with discovery and authentication of personal instruments. MEDIC system hosts IDM, which allows it to support local sensing, signal processing and autonomous decision making. MEDIC also supports remote configuration and control, which enables selective (de)activation of individual ADC sensing channels and general system configuration with GUI based administration tool at backend (central) server. MEDIC encompasses three network tiers (Fig. 3). The system relies on the established Internet network infrastructure in the first tier, and standard, ubiquitous wireless/GPRS in a second tier. The first two network tiers connect a patient's handheld personal server with servers at a hospital or clinic. Raw sensor data, patient events, and context can be transmitted for storage to centralized servers, or streamed for real-time monitoring of the patient's condition. MEDIC's system hardware is based on standard commercial components that offer a path for low-cost and ubiquitous deployment, rapid system integration during development, as well as ready adoption by collaborators. An additional benefit is the possibility for the hardware to improve with time by taking advantage of the rapid evolution in consumer products. In contrast to the systems in Section 2, the MEDIC system offers an integrated approach based on standard hardware and software, with emphasis on local signal processing, wireless sensing, high-bandwidth data streaming from multiple wireless sensing devices, as well as interaction. Most importantly, MEDIC adapts its use of sensors to optimize the accuracy of diagnostic inference regarding patient condition (Section 3).

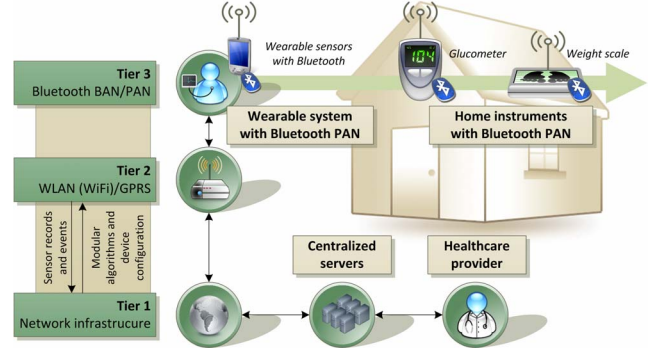


Figure 3. MEDIC's three-tier system architecture.

4.1 Hardware Architecture

Broadly, MEDIC uses a three-tier framework (Fig. 3), similar to [32, 36], with: sensors (wearable and instruments), personal servers (i.e., handheld device), and a central server (i.e., medical gateway), offering a flexible and high-capacity acquisition of various biometric data. Each tier is described briefly as follows:

4.1.1 Sensor tier

Sensors consist of different wearable sensors and compact, low-power Bluetooth boards [4] with low power radios and analog-to-digital converters. Initial sensors include ECG, accelerometers, gyroscopes, and knee angulation [8, 1, 2] (Fig. 4). These Bluetooth sensors can acquire data up to 1 kHz of 16-bit data from one channel or up to 300 Hz of 16-bit data from all eight ADC channels. In addition to these body-worn sensors, the prototype system includes standalone instruments such as a weight scale, blood pressure cuff, and blood oximeter [6, 5, 7], which already have Bluetooth radios or can be equipped with a serial-Bluetooth adapter [3].

4.1.2 Personal server/handheld tier

The handheld personal server is a programmable lightweight system, such as a cell phone or PDA. These devices serve as on-body terminals as a means of data collection from sensors, local data processing and transmission to enterprise. The prototype Medic platform is based on a standard Nokia N770 PDA, although any standard cell phone or PDA equipped with Bluetooth could be used. A wireless body area network (WBAN) of wearable sensors can thus be established via Bluetooth; the PDA in turn connects to enterprise servers (i.e., the central server tier) with a standard WiFi protocol. Bluetooth

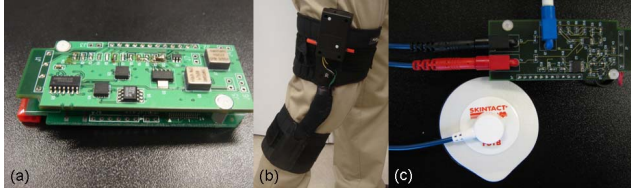


Figure 4. MEDIC's wearable sensors. (a) accelerometer and gyroscope, (b) knee angulation sensor, and (c) ECG.

wireless capacity on the personal server will thus be used to connect to these local non-wearable instruments in the form of a wireless personal area network (WPAN).

4.1.3 Central server/medical gateway tier

The third layer of the MEDIC system provides broadband connectivity from the handheld to the medical enterprise.

4.2 Software architecture

The main part of the software (Fig. 5) is a multi-threaded device server that facilitates the connection between wireless sensing devices and client services, such as local signal processing, a graphical user interface (GUI) for patient interaction and data logging. These individual client programs are connected to each other and the device server via TCP/IP sockets, enabling reliable, language-independent communication across heterogeneous hardware in different locations. Through this network abstraction, other clients located on centralized servers can receive patient data in real-time. More importantly, users may connect to personal server and seamlessly monitor data in real-time or take control of the diagnostic processes on the personal server, if desired. The main components of the software architecture are described briefly as follow:

4.2.1 Device server

The device server enables connection between the wireless sensors and client programs. Communication between clients and sensor driver threads is handled asynchronously via dedicated buffers associated with each sensing device. The client interface threads retrieve incoming data from each sensor buffer and forward it to the corresponding client programs by pushing data over TCP/IP sockets.

4.2.2 Sensing device drivers

A device driver thread for the wireless sensors consists of a state machine that receives configuration commands from

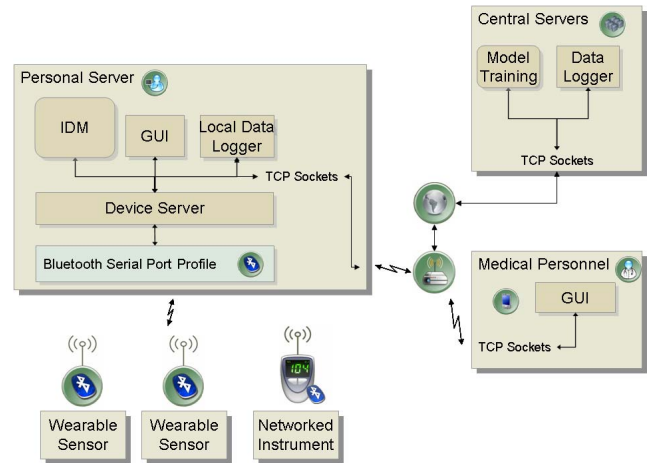


Figure 5. MEDIC's software architecture.

its command buffer and writes the incoming data to its data buffer. The device driver thread communicates with the wireless sensor via the Bluetooth Serial Port Profile (BSPP), emulating serial protocol communication. The commands include reconfiguration, adaptive sampling, and signal processing parameters (e.g., ADC sampling rate, number of ADC channels, start sampling, stop sampling, filter coefficients) for data streaming devices. In addition, data from existing standalone home instruments using RS-232-based serial protocols can be sent.

4.2.3 Clients

Because of the language-independence offered by TCP/IP interfaces included in this architecture, the client programs can be written in multiple programming language choices: C/C++, Java, Python, etc. Dedicated liaison threads facilitate the flow of sensor data and commands between the device server and the client programs. Client liaison threads are also simple state machines that read data from the corresponding sensor buffers and write commands into the command buffers. Simultaneous command writes are prevented and data coherency is ensured by individual mutex locks implemented on the buffers.

5 Preliminary Results

A pilot study of this hierarchical sensor network using Bayesian statistical inference was conducted for limp detection and characterization [79]. Two different types of sensors were used in this early investigation to collect training data: 6 accelerometers (left/right, x/y/z axes) on each hip; and 2 knee braces fitted with flex sensors to measure knee angulations. The accelerometers provided low-level motion sensing; if the probability of classification of limp

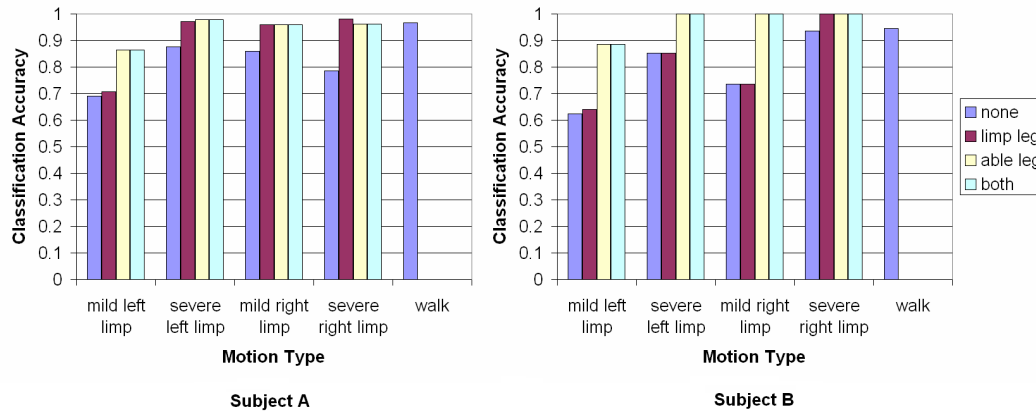


Figure 6. Detection accuracy of the system in distinguishing different degrees of limp and walk for two subjects when accelerometers on both sides of the hip are used and no knee sensor is used, a knee sensor is used on the limp leg, a knee sensor is used on the able leg, or both knee sensors are used. The results are normalized.

from these sensors did not meet a given threshold (e.g., probability of certainty was $<95\%$), and/or a more specific limp classification was required (e.g., is the limp due to a shortened leg, stiff knee, or antalgic?), additional sensors were activated (e.g., the knee brace) to provide more information to confirm the classifier decision. Preliminary evaluation of this hierarchical approach successfully showed that the classifier, using only the accelerometers (without any knee sensors), had lower accuracy than when the system includes the knee sensors, as shown in Fig. 6. Four human subjects were asked to limp/walk in a rectangular building with a long (50 meters) hallway on each side of the building, wearing both knee sensors, two accelerometer units, shoe inserts to simulate severity levels of limping, and a prototype wearable system [79]. We collected a total of 300 sets of experimental data from each of the four human subjects (1200 sets total) for training another 50 sets each for evaluation. For each of the subjects the data includes 50 sets for each of the four limp types, 50 sets for walking and another 50 sets for stationary state (sensor initialization or default state). The data collection of 50 sets corresponds to 15 minutes of walk/limp time.

As a part of the MEDIC system, an inference engine for gait analysis was created for real-time patient monitoring by two triaxial accelerometers located on each hip. Voltage signals from each triaxial accelerometer were sampled at 100 Hz and digitized into 16-bit data, which was then transferred in sequence over the Bluetooth WBAN to the Nokia PDA. The personal server in turn transmits the data via network socket interfaces to the corresponding graphical user interface (GUI) displays on both the PDA and a centralized server; Fig. 7 shows, the data streamed in real-time from the wearable system to a centralized server over Wi-Fi. Along

with the raw data, healthcare personnel can view the diagnosis results and status of the wearable system.

6 Conclusion & Future Works

In this paper, we described a novel decision support system, the Incremental Diagnosis Method (IDM) [79] based on Bayesian statistics and decision analysis theory. This is developed to address the fundamental requirements for methods that systematically guide the selection and activation of patient wearable sensors. The selection of sensors is based on direct computation of both the resource cost and the contribution of each sensor to the certainty of diagnosis, providing a framework to discern when to use available sensors to improve diagnostic certainty at various resolution levels of patient condition. We also described a wearable testbed system based on standard commercial systems. This testbed, Medical Embedded Device for Individualized Care (MEDIC), hosts IDM for local signal processing and decision support. We also showed that the testbed serves as a gateway for real-time health monitoring with body-worn sensors attached to Bluetooth devices in a wireless body area network. This paper also described an autonomous IDM solution being adopted for gait analysis in rehabilitative medicine. Both the IDM system and the MEDIC testbed are available as open source shared systems for research. Research opportunities in the future and those currently being pursued include integration of many diverse sensor and data sources, and many IDM applications in various patient environments with objectives in disease detection, rehabilitation, therapy, and wellness management. An important application being studied is chronic obstructive



Figure 7. Real-time patient monitoring and sensor reconfiguration in MEDIC.

pulmonary disease (COPD), which requires the correlation of many different patient parameters and benefits directly from the methods described here.

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